

BUILDING CHORDS FROM STACKED THIRDS

Every chord quality is determined by which thirds are stacked. Two intervals create a triad; a third interval creates a 7th chord.

CHORD FORMULAS AS STACKED INTERVALS

Major 3rd (M3) = 4 semitones

Minor 3rd (m3) = 3 semitones

Root note

Major C (C-E-G)	C + M3 (C → E) + m3 (E → G) M3 + m3
Minor Cm (C-E $\flat$ -G)	C + m3 (C → E$\flat$) + M3 (E$\flat$ → G) m3 + M3
Diminished C $^\circ$ (C-E $\flat$ -G $\flat$)	C + m3 (C → E$\flat$) + m3 (E$\flat$ → G$\flat$) m3 + m3
Augmented C+ (C-E-G $\sharp$)	C + M3 (C → E) + M3 (E → G$\sharp$) M3 + M3
Dom7 C7 (C-E-G-B $\flat$)	C + M3 + m3 + m3 ($\flat$7)
Maj7 C Δ 7 (C-E-G-B)	C + M3 + m3 + M3 (maj7)
Min7 Cm7 (C-E $\flat$ -G-B $\flat$)	C + m3 + M3 + m3 ($\flat$7)

DIM7 SYMMETRY

C $^\circ$ 7: 3 stacked minor 3rds

m3 (G $\flat$ → A = dim7)

m3 (E $\flat$ → G $\flat$)

m3 (C → E $\flat$)

C

All 4 notes are equal. Any one can be "the root" — divides the octave into 4 equal parts.

ONLY 3 UNIQUE DIM7 SOUNDS

C – E $\flat$ – G $\flat$ – A
= 4 inversions of the same sound

C $\sharp$ – E – G – B $\flat$
= 4 inversions of the same sound

D – F – A $\flat$ – B
= 4 inversions of the same sound

The other 9 "dim7 chords" you might name are just inversions of these 3 sounds.